31 ☆ QUESTION: 31 GROUP: **Technical** SECTION: **AI Fluency** Mark(s): 1

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Why can metadata filtering improve vector retrieval results?

- It limits similarity search to relevant structured conditions
- It increases the number of dimensions in every embedding
- It removes the need to calculate similarity between vectors
- It changes all stored vectors into identical representations

QUESTION: **32** GROUP: **Technical** SECTION: **SQL** Mark(s): 1

Consider the Employee table given below:

Employee				
Eno	**Ename**	**Job**	**Comm**	**sal**
7788	Raj	Manager	200	8000
7369	Ram	Salesman	100	5000
7521	Deepak	Analyst	50	7000
7892	Rahul	Manager	300	9000

Which one of the query given in options will display the employee details for the person who is earning third least salary?

- SELECT * from Employee e where 4=(select count(*) from Employee a where e.sal > a.sal);
- SELECT * from Employee e where 2=(select count(*) from Employee a where e.sal > a.sal);
- SELECT * from Employee e where 3=(select count(*) from Employee a where e.sal > a.sal);
- SELECT * from Employee e where 2=(select count(*) from Employee a where e.sal < a.sal);

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QUESTION: **33** GROUP: **Technical** SECTION: **Cloud Computing Basics** Mark(s): 1

What are the advantages of data virtualization?
I) It reduces costs by removing data replication.
II) It increases business user efficiency by providing data in real-time.
III) It creates availability issues, because availability is maintained by third-party providers.
IV) It allows users to access the data without worrying about where it resides on the memory.

- Only(I),(II) and (III)
- All of the options given.
- Only (I),(II) and (IV)
- Only (I)

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QUESTION: **35** GROUP: **Technical** SECTION: **Cloud Computing Basics** Mark(s): 1

The role of hypervisor is/are:

- automatically expanding or compressing the infrastructural resources on a sudden-up and down in the requirement.
- creates and runs virtual machines and allows multiple Operating Systems to share a single physical hardware.
- Both I and II
- Neither I nor II

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32 What will be the output of the following code?

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```
class Base {
public:
    virtual void show() { cout << "Base"; }
};
```
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```
class Derived : public Base {
public:
    void show() override { cout << "Derived"; }
};
```
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```
int main() {
    Base* b = new Derived();
    b->show();
    return 0;
}
```
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- ○ Base
- ○ Runtime Error
- ○ Compiler Error
- ○ Derived

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What will be the output of the program given below?

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```
public class Sample
{
    int x=10;
    public void m1()
    {   System.out.print("A ");   }
    public class ReallyInner
    {
        public void m1()
        {   System.out.print("B ");   }
        public void test()
        {
            this.m1();
        Sample.this.m1();
    }
}
public static void main(String... args)
    {
    Sample.ReallyInner i = new Sample().new ReallyInner();
    i.test();
    }
}
```

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Which prompting approach can help when the model lacks enough information?

- Instruct it to create plausible details whenever evidence is missing
- Instruct it to maximize creativity for every factual response
- Instruct it to state uncertainty instead of inventing missing details
- Instruct it to ignore all retrieved sources before answering

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QUESTION: 43 GROUP: Technical SECTION: AI Fluency Mark(s): 1

A team is choosing between an RNN and a Transformer for processing long sequences. Training throughput on modern parallel hardware is a major concern. Why can a Transformer provide an advantage during training?

- Transformers remove matrix multiplications and therefore require fewer numerical operations than recurrent neural networks.
- Self-attention always has linear complexity with sequence length, making it cheaper than recurrence for every sequence size.
- Transformers maintain only one hidden state, eliminating intermediate activations that must be retained for backpropagation.
- Self-attention allows sequence positions to be processed with greater parallelism than strictly sequential recurrent updates.

A program stores the marks of students in a test. It performs some updates on the list and prints the final average score.

```
marks ← [32, 45, 28, 50, 60]
FOR i ← 1 TO LENGTH(marks)
    IF marks[i] < 40 THEN
        marks[i] ← marks[i] + 5
    END IF
END FOR
sum ← 0
FOR i ← 1 TO LENGTH(marks)
    sum ← sum + marks[i]
END FOR
PRINT sum / LENGTH(marks)
```

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A recruiting model is trained primarily on historical hiring decisions from a period when some groups were underrepresented in selected candidates. What is the main Responsible AI risk?

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43 Removing candidate names guarantees that historical representation bias cannot influence predictions through other correlated features.

44 Historical selection patterns may be learned and reproduced, creating or amplifying existing representation and outcome biases.

45 ☆ Underrepresented groups automatically receive higher model uncertainty, which guarantees conservative and therefore fair hiring decisions.

46 Historical data prevents overfitting because past human decisions provide naturally balanced labels for every demographic population.

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Rishita has written an HTML code where she wants to replace the text "Central Style Sheets" with "Cascading Style Sheets" mentioning the time of change. She also wants to insert a link to provide information regarding the change. She is facing some problems. Analyze Rishita's code, find the errors and suggest the changes.

```
<body>
  <p>CSS stands for <del href="https://css.com/research" date="2018-11-21T15:55:03Z">Central Style Sheets</del>
<ins>Cascading Style Sheets</ins></p>
</body>
```

- The "href" attribute should be replaced with "link" attribute
- The "href" attribute should be replaced with "cite" attribute
  The "date" attribute should be replaced with "datetime" attribute
- The "href" attribute should be replaced with "cite" attribute
  "<ins>" tag should be replaced with "<add>"
- The "date" attribute should be replaced with "datetime" attribute
  "<ins>" tag should be replaced with "<add>"

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Consider the table given below:

HOUSE_RENT		
HID	HSIZE	RENT
4562	1BHK	9500
4563	2BHK	13500
4567	3BHK	18000
4568	1BHK	11000

Which of the option given below query will add not null constraint to the hsize column of HOUSE_RENT table?

I) `Alter table HOUSE_RENT modify hsize not null;`

II) `Alter table HOUSE_RENT add not null (hsize);`

III) `Alter table HOUSE_RENT add constraint not null (hsize);`

- Only (I)
- Only (III)
- All (I), (II) and (III)
- Only (II) and (III)

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The primary purpose of a validation dataset during model development is to:

49 tune model choices before final evaluation

50 store predictions after deployment

51 increase the number of input features

52 replace the training dataset completely

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The user is transmitting some data across a network by considering the below given features.

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I) Packet sequencing
II) Addressing
50 III) Congestion control

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In which of the given layer, the transmission might be going on?

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- Data link layer

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- Transport layer

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- Presentation layer

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- Network layer

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